

INTERNAL GUIDE · BUSINESS DEVELOPMENT TEAM

How we build the North Park 3D Home Experience

Software, equipment, people, process and costs, explained simply.

Project	The North Park, Shantigram, Ahmedabad (Adani Realty)
Prepared by	Applore Technologies
Date	November 2026
For	Internal use. Not for sharing with the client as is.

Contents

01 The idea in one page

02 Words you will hear

03 How it is made, step by step

04 Software we use

05 Equipment

06 People and who does what

07 What it costs to build

08 What it costs to run

09 Cost per homeowner

10 Packages we can offer

11 How the work runs after launch

12 What we need from Adani Realty

13 Summary

All prices are approximate, collected in November 2026. Dollar prices are converted at ₹85 per US dollar and ₹92 per euro. Costs in this guide are what it costs us to make and run the product, not the price we quote to the client.

01

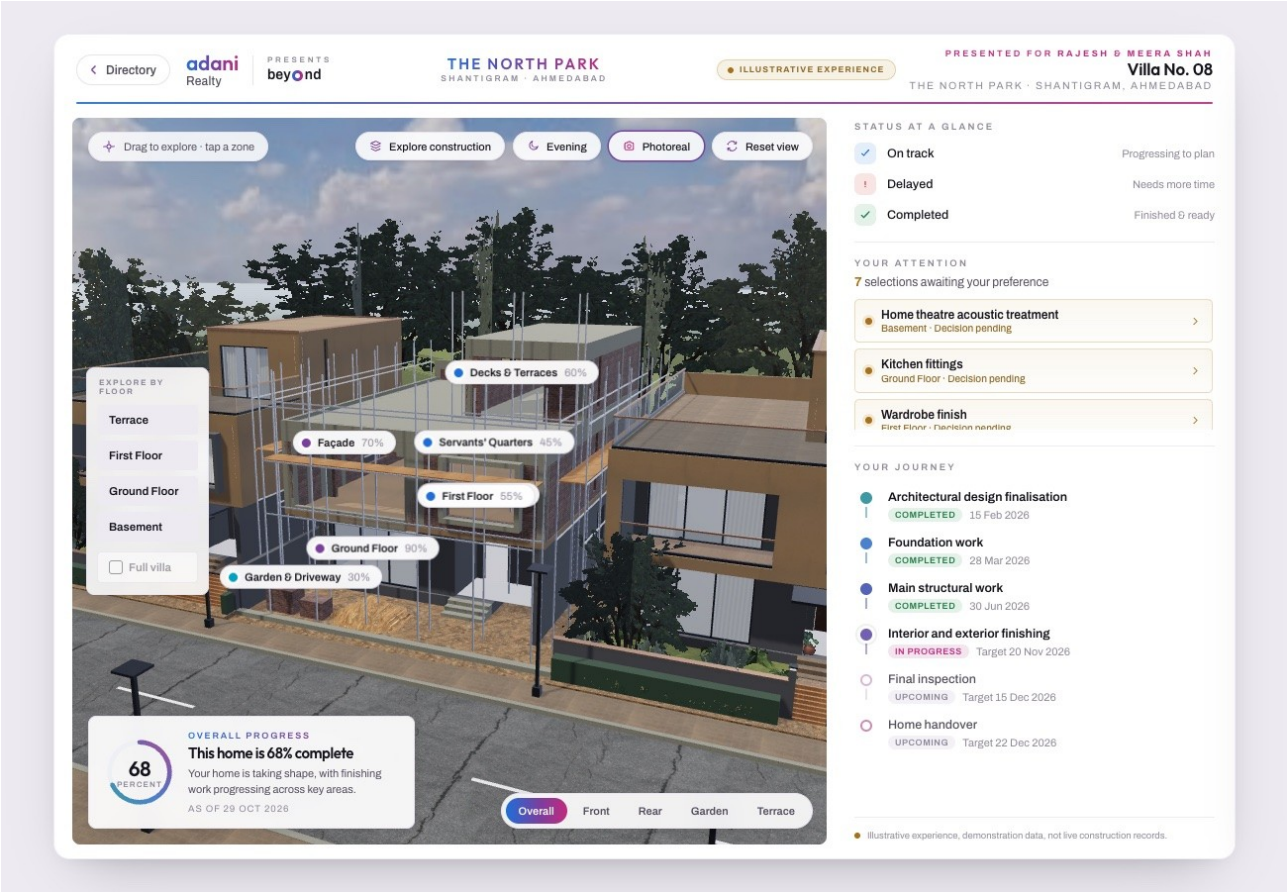
The idea in one page

Every North Park homeowner gets their own villa in 3D, on their phone or laptop. The model shows how far construction has reached today, what the finished home and its interiors will look like, and real photos from site every month.

We produce three things

1. **3D models** of each villa type, outside and inside, in every construction stage.
2. **Photo-quality images and short films** made from those models.
3. **The website and app** where homeowners, Relationship Managers and site teams see and update everything.

> ### The one thing to remember
> We build one model per villa TYPE, not one per homeowner. The North Park has three types (4, 5 and 6 BHK).
>
> Every homeowner's villa is the same model with their own data: their construction stage and their chosen finishes. So the big cost is paid once, and each extra homeowner costs very little.



Our working prototype, live today: Villa No. 08 shown at its real construction stage, with pending decisions and milestones on the right.

02

Words you will hear

A short dictionary for anyone new to the project.

Word	What it means
3D model	A digital version of the villa that you can turn around, look into and walk through on a screen.
Exterior / interior	The outside of the villa (walls, balconies, garden) and the inside (rooms, furniture, finishes).
Render	A picture made by the computer from the 3D model. A photoreal render looks like a real photograph.
Construction stage	Where a part of the home is: Foundation, Structure (concrete and bricks), Finishing (plaster and windows) or Ready.
Texture	The surface look of a material, such as wood grain, stone or plaster, applied to the model.
Sky image (HDRI)	A special 360° photo of the sky that lights the 3D scene so it looks like real daylight or evening.
Finishes / selections	Choices a homeowner makes: flooring, kitchen, wardrobes, bathrooms, doors and so on.
3D viewer	The part of the website or app that shows the 3D model and lets people move around it.
Hosting	Renting servers on the internet so the website and app are always online.
Drone and 360° capture	Monthly aerial photos of the site from a drone, and 360° photos walking through each villa.
Licence	The yearly fee to use a paid software.
Cloud GPU	A powerful graphics computer rented by the hour on the internet, used to make many images quickly.

03

How it is made, step by step

Eight steps. Steps 1 to 6 happen once per villa type; steps 7 and 8 repeat every month.

Step 1. Collect drawings and finishes

Floor plans, elevations and sections of each villa type, the architect’s 3D files if they exist, and the finishes catalogue.

Who	BD team with Adani Realty
Tools	Email, shared drive
Time	1 to 2 weeks

Step 2. Build the outside of each villa type

The artist builds walls, windows, balconies, terrace, gate and garden to exact size from the drawings.

Who	3D artist
Tools	Blender (free); SketchUp only to open the architect’s files
Time	1 to 2 weeks per type

Step 3. Build the inside

Every room is furnished and lit. Each finish option (for example three flooring choices) is prepared so it can be swapped.

Who	Interior 3D artist
Tools	Blender, free texture libraries, paid furniture models when needed
Time	2 to 4 weeks per type

Step 4. Prepare the construction stages

The script turns one finished model into its Foundation, Structure, Finishing and Ready versions automatically.

Who	Our developer
Tools	Our own Blender script (already built)
Time	2 to 3 days per type

Step 5. Make photo-quality images and films

Images of each home at its stage, in day and evening light. Handover films at the end.

Who	Automatic, checked by the 3D artist
Tools	Blender (free) on our Mac; D5 Render for quick marketing videos
Time	Minutes per image

Step 6. Put it in the website and app

The models are prepared for phones and laptops, then shown to each homeowner with their own data.

Who	Developer and designer
Tools	Babylon.js 3D viewer (free), our app, Vercel hosting
Time	Part of the app build

Step 7. Capture the real site every month

Aerial photos of the whole development and 360° photos of each villa under construction.

Who	Drone pilot (vendor or trained staff)
Tools	DJI drone, Insta360 360° camera
Time	1 day per month

Step 8. Update progress every month

Engineers mark each part of each villa. The app updates the 3D home, makes new images and notifies the homeowner.

Who	Adani site engineers, approved by their project manager
Tools	Our site progress console on a tablet
Time	Ongoing

04

Software we use

Most of what we need is free. Paid software is only added where it saves real time.

Core software

Software	What it does	Needed?	Cost
Blender	Builds the 3D villas and interiors, and makes the photo-quality images	Must	Free
Babylon.js	Shows the 3D villa inside the website and app	Must	Free
Poly Haven	Free library of textures, skies and plants used in our models	Must	Free
Our Blender script	Turns each model into all construction stages and makes images automatically	Must	Built by us
Vercel Pro	Hosts the website so it is always online	Must	$20 per seat per month (about ₹ 1,700)
GitHub	Stores our code safely, with full history	Must	Free for our team size
Apple developer account	Needed to publish the iPhone app	For the app	$99 (about ₹ 8,400) per year
Google Play account	Needed to publish the Android app	For the app	$25 (about ₹ 2,100) one time

Paid software we may add

Software	What it does	Needed?	Cost per year
D5 Render Pro	Makes marketing images and walk-through videos quickly	Recommended	$360 (about ₹ 30,600)
SketchUp Pro	Only to open the architect's files if they use SketchUp	If needed	$399 (about ₹ 33,900)
Paid furniture models (TurboSquid, CGTrader)	Ready-made sofas, beds, lights and decor for interiors	Sometimes	About ₹1,500 to ₹8,000 per item (estimate)
Lumion Pro	Alternative to D5. Runs only on Windows	Not needed	$1,149 (about ₹ 97,700)

Software	What it does	Needed?	Cost per year
V-Ray or Corona	Used by outside studios that work in 3ds Max; they pay for it	Studio pays	$395 (about ₹ 33,600) to $540 (about ₹ 45,900)
Cloud GPU (RunPod, AWS)	Rented graphics computers to make many images fast once we have hundreds of homes	Later	$0.34 to $0.80 per hour (about ₹30 to ₹70)
Unreal Engine with cloud streaming	Premium, game-like interiors streamed to the homeowner's browser	Premium option only	About €9 per viewing hour per user (about ₹ 828), plus platform fees

Why we chose free tools first

Blender and Babylon.js are used by large studios worldwide and cost nothing. They let us own the whole process, change anything quickly, and keep the per-homeowner cost very low. Paid tools are added only when they clearly save time.

05

Equipment

Item	Why we need it	Cost
Mac with Apple Silicon	Runs Blender, makes images, runs the image server. We already have it.	₹0 (already owned)
DJI Mini 4 Pro drone	Monthly aerial photos of the development	₹95,000 to ₹1,05,000
Insta360 X5 360° camera	Monthly 360° walk-throughs of each villa	₹55,000 to ₹59,000
Windows PC with a strong graphics card	Only if we choose Lumion or Unreal later	₹3 to 4 lakh (estimate)
Tablets for site engineers	To update progress from site (usually Adani's)	₹30,000 to ₹50,000 each
Sales gallery screen and headsets	Only if Adani wants the gallery experience	₹7 to 9.5 lakh (estimate)

Instead of buying the drone and camera, we can hire a local drone and 360° vendor for about ₹40,000 to ₹60,000 per month (estimate). Buying is cheaper after about four months, but needs a trained pilot and drone permissions.

06

People and who does what

Freelance 3D artists in India commonly charge $10 to $25 per hour; experienced archviz artists on contract earn about ₹30,000 to ₹90,000 per month.

Role	What they do	Typical cost
Exterior 3D artist	Builds the outside of each villa type from the drawings	₹30,000 to ₹90,000 per month, or per villa type
Interior 3D artist	Builds and furnishes every room, prepares finish options	₹30,000 to ₹90,000 per month, or per villa type
Developer	Website, app, 3D viewer, automation and integrations	Our team
Designer	Screens and look of the app	Our team
Project manager	Plans, tracks and talks to Adani Realty	Our team
Drone pilot	Monthly site capture	Vendor, or trained staff member
Adani site engineers	Update progress every month	Adani Realty’s team

07

What it costs to build

One-time production cost for the three North Park villa types. The range is wide because a freelancer and a large studio can differ three to four times in price.

Item	Per villa type	All three types
Outside model (from drawings)	₹40,000 to ₹1,50,000	₹1.2 to 4.5 lakh
Inside model, all rooms furnished	₹1,50,000 to ₹5,00,000	₹4.5 to 15 lakh
Construction stages	₹10,000 to ₹20,000	₹0.3 to 0.6 lakh
Finishes library (about 120 options)	Shared	₹0.5 to 1.5 lakh
Total models		**₹6.5 to 21.6 lakh**

How to bring this cost down

If the architect shares their 3D files, modelling cost falls by 50 to 70%, because the artist starts from a finished shape instead of drawing from scratch.

Start with one villa type (4 BHK), show it to homeowners and Adani leadership, then build the other two.

The website and app (homeowner app, RM portal, site console, configurator and dashboard) are built by our own team. Their cost depends on scope and is quoted separately in the client proposal.

08

What it costs to run

Running costs after launch, per month. The middle figure is what we use for the per-homeowner calculation.

Item	Range	We assume
Hosting (Vercel Pro, two seats)	₹ 3,400 plus usage	₹ 3,400
Software licences (D5, SketchUp, Apple)	₹ 6,100 averaged per month	₹ 6,100
Drone and 360° capture (vendor)	₹40,000 to ₹60,000	₹ 50,000
Support and small updates (part-time developer)	₹50,000 to ₹1,00,000	₹ 75,000
Fixed monthly total		**₹ 1,34,500**

Costs that grow with each homeowner

Item	How it is worked out	Per homeowner per month
New images each month	About 6 images, 2 minutes each on a rented cloud GPU	₹10
Storage and data	Photos, 360° images and documents	₹5
Total		**₹ 15**

If we keep making images on our own Mac, the ₹10 for images becomes almost zero. A cloud GPU is only needed when there are hundreds of homeowners and images must be ready quickly.

09

Cost per homeowner

Most costs are shared, so the more homeowners use the platform, the cheaper it is per homeowner.

	100 homeowners	300 homeowners	500 homeowners
Running cost per homeowner per month	₹ 1,360	₹ 460	₹ 280
Running cost per homeowner per year	₹ 16,320	₹ 5,520	₹ 3,360
Build cost shared per homeowner (one time, at ₹14,00,000 for models)	₹ 14,000	₹ 4,700	₹ 2,800

₹ 460
per homeowner per month at 300 homeowners

₹ 4,700
one-time model cost per homeowner at 300

₹ 0
extra model cost for each new homeowner

Premium option: game-like interiors

If Adani Realty wants homeowners to walk through photo-real interiors live in the browser, we stream them from cloud computers. This costs about €9 (about ₹ 828) per hour of viewing. A homeowner watching two hours a month adds about ₹ 1,700 per month. We recommend this only for the sales gallery or as a paid upgrade.

10

Packages we can offer

Three levels, so the client can start small and grow.

	Basic	Standard	Premium
3D outside of each villa type	Yes	Yes	Yes
Construction stages	Yes	Yes	Yes
Monthly site photos and progress	Yes	Yes	Yes
Photo-quality images	Yes	Yes	Yes, plus films
Furnished interiors	No	Yes	Yes
Design Your Home (finishes)	No	Yes	Yes
Game-like live interiors	No	No	Yes
Sales gallery screen and headsets	No	Optional	Yes
Model cost, three types	₹1.5 to 5 lakh	₹6.5 to 21.6 lakh	₹8 to 25 lakh plus streaming

11

How the work runs after launch

Every month follows the same rhythm.

When	What happens
Week 1	Drone flight over the development and 360° walk-through of each villa under construction.
Week 1 to 2	Site engineers mark the progress of each part of each villa on the tablet and add photos.
Week 2	Adani's project manager approves the updates.
Week 3	The system moves each 3D home to its new stage, makes new images and prepares the monthly update.
Week 3	Homeowners get a notification; Relationship Managers see who has viewed it and what is pending.
Any time	New finish options, fixes and small improvements by our support developer.
Every year	Renew software licences, app store accounts and hosting; review costs with Adani Realty.

12

What we need from Adani Realty

A checklist for the BD team. The first two items decide our modelling cost and timeline.

1. **Drawings for each villa type:** floor plans with sizes for every floor (basement to terrace), elevations and sections, as PDF or DWG.
2. **The architect's 3D files, if they exist:** SketchUp, Revit, 3ds Max or FBX. This saves the most money.
3. **Finishes catalogue:** every option offered to homeowners, with photos, brand names and any price difference.
4. **Villa list:** each villa number, its type and its homeowner.
5. **Construction schedule** for each villa, and a site contact.
6. **Site access and drone permission** for monthly capture.
7. **Brand guidelines** and approval of how Adani Realty appears in the app.
8. **One decision-maker** on the Adani Realty side for reviews and sign-off.

13

Summary

Question	Answer
Which software makes the 3D villas and interiors?	Blender (free)
Which software makes the photo-quality images?	Blender (free); D5 Render for quick videos
What shows the 3D villa in the app?	Babylon.js (free)
What do we need to buy?	Drone and 360° camera, or hire a vendor
One-time model cost, three villa types	₹6.5 to 21.6 lakh
Fixed running cost per month	₹ 1,34,500
Cost per homeowner per month (300 homeowners)	₹ 460
Extra model cost for each new homeowner	₹0
What lowers cost the most?	Getting the architect’s 3D files

Sources for prices

D5 Render: g2.com, softwareadvice.com

Lumion: myarchitectai.com, renderahouse.com

V-Ray, Corona, Enscape: renderahouse.com, cgchannel.com

SketchUp: myarchitectai.com, renderahouse.com

Vercel: vercel.com/docs/plans/pro, makerkit.dev

Cloud GPU: getdeploying.com, computeprices.com

Pixel streaming: vagon.io comparison of Arcware

DJI Mini 4 Pro and Insta360 X5: xboom.in, smartprix.com

Rendering and artist rates in India: peninsulartworks.com, indeed.com, freelancer.com